

Communication Electronics Cheat Sheet (Equations + Notes)

1. Waves & Antennas

- Wavelength:
$$\lambda = \frac{c}{f}$$
- Quarter-wave antenna:
$$L = \lambda/4$$
- Half-wave dipole:
$$L = \lambda/2$$
- Propagation time:
$$t = d/v_p$$
- Velocity in coax:
$$v = c \cdot VF$$

Notes:

- Quarter-wave antennas reflect from the ground to appear as half-wave.
- Coax has velocity factor < 1, reducing propagation speed.
- Free space propagation delay can be significant for planetary distances (Mars-Earth ~12.5 min).

2. Decibels & Gains

- Voltage gain (dB):
$$G_v(dB) = 20 \log_{10}(V_{out}/V_{in})$$
- Current gain (dB):
$$G_i(dB) = 20 \log_{10}(I_{out}/I_{in})$$
- Power gain (dB):
$$G_p(dB) = 10 \log_{10}(P_{out}/P_{in})$$
- Cascaded gains:
$$G_{tot} = \prod A_n, \quad G_{dB} = \sum G_{n,dB}$$
- Power in dBm:
$$P_{dBm} = 10 \log_{10}(P/1mW)$$

Notes:

- Loss is expressed as negative dB gain.
- Use 20 log for V and I ratios, 10 log for power.
- Cascaded stages multiply numerically but add in dB.

3. RC Filters

- Corner frequency:
$$f_c = \frac{1}{2\pi RC}$$
- Attenuation slopes:
 - Single pole: -20 dB/decade (−6 dB/octave)
 - Two pole: -40 dB/decade (−12 dB/octave)
 - Transfer function:
$$H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$$
- At corner frequency:
$$|H(j\omega_c)| = 1/\sqrt{2} = -3dB$$

Notes:

- LPF passes low frequencies, attenuates high.
- HPF passes high frequencies, attenuates low.
- Attenuation slope depends on number of poles.

4. Resonance & RLC Circuits

- Inductive reactance:
$$X_L = 2\pi fL$$
- Capacitive reactance:
$$X_C = \frac{1}{2\pi fC}$$
- Resonant frequency:
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$
- Quality factor:
$$Q = \frac{X_L}{R_s} = \frac{f_0}{BW}$$
- Bandwidth:
$$BW = f_0/Q$$

Notes:

- At resonance, $X_L = X_C$.
- Tank circuits: high impedance at resonance, block current.
- Higher Q = narrower bandwidth.
- Inductor Q:

$$Q = X_L/R_s$$

$$Q = R_p/X_C$$

5. LC Filters

- Butterworth (maximally flat):

$$L_{ref} = \frac{R_0}{2\pi f_c}, \quad C_{ref} = \frac{1}{2\pi R_0 f_c}$$

- Chebyshev (ripple, sharp cutoff):

$$L_{ref} = \frac{R_0}{2\pi f_c}, \quad C_{ref} = \frac{1}{2\pi R_0 f_c}$$

Notes:

- Insertion loss: signal loss even in passband.
- Transition region: between stopband and passband.
- Butterworth = smooth response, Chebyshev = sharper roll-off with ripple.

6. Sallen-Key Filters

- Component relations:

$$C_2 = 2C_1$$
$$R_1 = R_2 = \frac{1}{2\pi f_c C_1 \sqrt{2}}$$

Notes:

- Active filter topology for implementing 2nd-order LPF.

7. Matching Networks

- Quality factor:

$$Q = X_L/R_s$$

- Equivalent parallel resistance:

$$R_p = (Q^2 + 1)R_s$$

- Parallel inductance:

$$L_p = L_s(1 + 1/Q^2)$$

- Parallel capacitance:

$$C_p = 1/((2\pi f)^2 L_p)$$

Notes:

- Shunt component sits on higher impedance side.
- Matching ensures max power transfer.
- At design frequency, series vs. parallel equivalents look identical.

8. Oscillators & AM

- Oscillator condition (Barkhausen):

$$|A\beta| \geq 1, \quad \angle A\beta = 0^\circ$$

- AM Signal:

$$s(t) = [1 + m \cos(\omega_m t)] \cos(\omega_c t)$$

- Modulation index:

$$m = A_m/A_c$$

Notes:

- AM = amplitude modulation, FM/PM = angle modulation.
- Need oscillator to generate carrier before modulation.
- Higher modulation index = deeper modulation, distortion if $m > 1$.